

	Case Name: Defense Building	Sector	Construction (institutional building)	
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources	
			Schedule with costs	
		Risk Analysis	Random simulation	
Submitted by	Mieke De Saedeleer		One of nine std. scenarios	
Date	2016		User defined distributions	
File Name	C2016-06 Defense Building	Project Control	Automatic tracking	
			Tracking based on user input	

1. Project description

Project authenticity

This project involves the construction and fit-out of a defense-related building with specialized technical spaces, including a cleanroom environment. It starts with site preparation, demolition, and structural works, followed by extensive installation of mechanical, electrical, piping, and HVAC systems. The project then focuses on interior finishing, cleanroom construction, and installation of equipment and control systems. Finally, it includes testing, validation (IQ/OQ/PQ), certification, and commissioning to ensure the facility meets strict operational and quality standards.

2. Project properties

2.1. Baseline Schedule

General		Network topology	
# Activities	98	Serial/Parallel (SP)	14%
Planned Duration (PD)	252	Activity Distribution (AD)	55%
Budget At Completion (BAC)	4331260.48	Length of Arcs (LA)	0%
Renewable Resources	-	Topological Float (TF)	76%
Consumable Resources	-		

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	9%	11%	3.98
CRI-rho	10%	12%	3.35
CRI-tau	9%	15%	4.80

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	N/A	N/A	N/A
CRI-rho	N/A	N/A	N/A
CRI-tau	N/A	N/A	N/A

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	11%	20%	1.33
SI	24%	26%	0.99
SSI	3%	7%	3.25
CRI-r	10%	9%	1.91
CRI-rho	10%	10%	2.10
CRI-tau	9%	15%	5.16

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	21%	-9%
PV - SPI	28%	20%
PV - SCI	28%	20%
ED - 1	51%	-7%
ED - SPI	28%	20%
ED - SCI	28%	20%
ES - 1	18%	-18%
ES - SPI(t)	26%	24%
ES - SCI(t)	26%	24%

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	0%	0%
CPI	0%	0%
SPI	13%	13%
SPI(t)	12%	12%
SCI	13%	13%
SCI(t)	12%	12%
0.8 CPI + 0.2 SPI	9%	9%
0.8 CPI + 0.2 SPI(t)	6%	6%

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 14 tracking periods. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	26.01	-642094.29	-37.61	0.93	0.65	0.60	0.93
std dev	97.32	443336.04	15.81	0.27	0.33	0.30	0.17
final	0.00	0.00	-20	1.00	1.00	0.93	1.00

2.3.3.2. Time forecasting

PD	252	Real Duration	232	Late/early	7% early
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EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	317.71	25.80	37%	-37%
PV - SPI	386.00	109.66	66%	-66%
PV - SCI	385.93	109.48	66%	-66%
ED - 1	310.36	18.50	34%	-34%
ED - SPI	387.43	108.29	67%	-67%
ED - SCI	387.36	108.12	67%	-67%
ES - 1	317.71	15.84	37%	-37%
ES - SPI(t)	505.64	292.79	118%	-118%
ES - SCI(t)	505.64	292.79	118%	-118%

2.3.3.3. Cost forecasting

BAC	4331260.48	Real Cost	4331260.48	Over/under budget	0
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EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	4331234.50	97.28	0%	0%
CPI	4331081.25	670.69	0%	0%
SPI	16739758.75	42058348.26	286%	-286%
SPI(t)	7303555.79	4707152.32	69%	-69%
SCI	16739388.89	42058416.23	286%	-286%
SCI(t)	7302942.86	4706011.88	69%	-69%
0.8 CPI + 0.2 SPI	4532123.43	288778.39	5%	-5%
0.8 CPI + 0.2 SPI(t)	4544881.50	268402.84	5%	-5%